Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, SQL, Java, C/C++, JavaScript, R

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming (OOP), System Design, Multithreading & Concurrency, Database Management Systems (DBMS), RESTful APIs

Data Science & Machine Learning: Machine Learning, Deep Learning (PyTorch), Statistics & Probability, Feature Engineering, Time-Series Analysis, Anomaly Detection, Pandas, NumPy, Scikit-learn

Generative AI: LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG)

Cloud & Tools: AWS, Google Cloud Platform (GCP), Git, GitHub, Docker, PyTest, CI/CD, Power BI

EXPERIENCE

VOIS – Vodafone Idea Foundation* |LINK** — ***Aug – Sep, 2025

- Collected and analyzed 2 real-world datasets (8,500+ records) using LLM-based prompt engineering, extracting 15 structured business insights through natural language querying.
- Iteratively refined prompts and validated AI-generated outputs for consistency and accuracy, improving reliability of generated insights.
- Built a repeatable data-processing pipeline integrating preprocessing, natural-language querying, response extraction, and validation, reducing manual steps in dataset analysis.

PROJECTS

***Real-Time Stock Market Anomaly Detector* |LINK**

- Designed and implemented an anomaly detection pipeline for NSE stock data, combining Isolation Forest with a PyTorch-based LSTM Autoencoder to flag abnormal price/volume movements from reconstruction error.
- Achieved ~88% precision and ~85% recall on a labeled validation set, benchmarked against a rule-based baseline.
- Integrated FinBERT for financial news sentiment scoring as an auxiliary signal; reduced false-positive alerts by ~25% versus the anomaly-only model.
- Performed feature engineering on time-series financial data, training and evaluating multiple models before selecting the best-performing configuration.

***RedisLite* |LINK**

- Engineered a Redis-inspired in-memory key-value database in Python featuring a custom RESP protocol parser, multithreaded TCP server, thread-safe storage engine, and an O(1) LRU cache with lazy/active TTL expiration.
- Designed a four-layer architecture (TCP transport, storage engine, cache/TTL, persistence) implementing Write-Ahead Logging (WAL), atomic log compaction, and automatic crash recovery.
- Validated the system with 46 automated PyTest unit tests (TDD), benchmarking 619K SET ops/sec in-memory and 2.8K ops/sec with durable WAL persistence.

***Customer Lifetime Value Engine* |LINK**

- Built a probabilistic ML pipeline using BG/NBD and Gamma-Gamma models on 1M+ historical transaction records to forecast customer lifetime value.
- Achieved a mean absolute percentage error (MAPE) of ~15% against actual 12-month customer value.
- Engineered features from raw transaction data and segmented customers into value tiers; identified the top 20% of customers driving ~65% of forecasted revenue.
- Visualized cohort-level trends in Power BI dashboards for stakeholder-facing reporting

***API Rate Limiter Library* |LINK**

- Built a production-ready API rate-limiting library in Python implementing five industry-standard throttling algorithms, Redis integration, and reusable middleware for Flask and FastAPI applications.
- -Implemented 30+ automated PyTest unit tests (TDD) and published the package on PyPI for public installation and reuse.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** **2026**
- Prepare Data for ML APIs on Google Cloud Skill Badge |**LINK** **2026**
- AWS Solutions Architecture Job Simulation |**LINK** **2025**